

PRACTICE PROBLEMS

1. Let $A = \begin{bmatrix} 1 & -1 & 5 \\ 2 & 0 & 7 \\ -3 & -5 & -3 \end{bmatrix}$ and $u = \begin{bmatrix} -7 \\ 3 \\ 2 \end{bmatrix}$. Is u in $\text{Nul } A$? Is u in $\text{Col } A$? Justify each answer.

Solution: By Defn, $\text{Nul } A$ represents the set of all vectors $x \in \mathbb{R}^3$ s.t. $Ax = 0$.

$$Au = \begin{bmatrix} 1 & -1 & 5 \\ 2 & 0 & 7 \\ -3 & -5 & -3 \end{bmatrix} \begin{bmatrix} -7 \\ 3 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \cdot (-7) + (-1) \cdot 3 + 5 \cdot 2 \\ 2 \cdot (-7) + 0 \cdot 3 + 7 \cdot 2 \\ (-3) \cdot (-7) + (-5) \cdot 3 + (-3) \cdot 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$\therefore u \in \text{Nul } A$.

By Defn, $\text{Col } A$ represents the set of all linear combinations of the columns of A . $A = [a_1 \ a_2 \ a_3]$.

$\therefore$ We need to verify whether there exists $c_1, c_2, c_3 \in \mathbb{R}$, s.t. $c_1 a_1 + c_2 a_2 + c_3 a_3 = u$. By Defn. of Ax , $[a_1 \ a_2 \ a_3] \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = u \Rightarrow Ax = u$.

By Theorem 3, from Chapter 1, the matrix equation $Ax = u$ has the same solution set as the system of equations corresponding to the augmented matrix $[A \ u]$.

$$[A \ u] = \left[\begin{array}{ccc|c} 1 & -1 & 5 & -7 \\ 2 & 0 & 7 & 3 \\ -3 & -5 & -3 & 2 \end{array} \right] \xrightarrow[\text{-2} \times \text{Row 1}]{\text{Row 2} = \text{Row 2}} \left[\begin{array}{ccc|c} 1 & -1 & 5 & -7 \\ 0 & 2 & -3 & 17 \\ -3 & -5 & -3 & 2 \end{array} \right] \xrightarrow[\text{+4} \times \text{Row 2}]{\text{Row 3} = \text{Row 3}} \left[\begin{array}{ccc|c} 1 & -1 & 5 & -7 \\ 0 & 2 & -3 & 17 \\ 0 & 0 & 0 & 49 \end{array} \right] \xrightarrow[\text{+5} \times \text{Row 1}]{\text{Row 3} = \text{Row 3}} \left[\begin{array}{ccc|c} 1 & -1 & 5 & -7 \\ 0 & 2 & -3 & 17 \\ 0 & 0 & 0 & 49 \end{array} \right]$$

It's easy to verify that the final matrix is in echelon form.

The 3rd row is of the form $[0 \ 0 \ 0 \ \dots \ 0 \ b]$ where $b \neq 0$. By Theorem 2 from Chapter 1, the system of equations is inconsistent, i.e. no solution set exists. $\therefore u \notin \text{Col } A$. $\square$

2. Given $A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$, find a vector in $\text{Nul } A$ and a vector in $\text{Col } A$.

Solution: By Defn, $\text{Nul } A$ represents the set of all vectors x s.t. $Ax = 0$. We know that $A \cdot 0 = 0$. $\therefore 0 \in \text{Nul } A$

By Defn, $\text{Col } A$ represents the set of all ~~vectors~~ linear combinations of the columns of A . $\text{Col } A = \left\{ c_1 \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + c_3 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\}$. $\therefore 0 \in \text{Col } A$. $\square$